ETALE COHOMOLOGY OF A QUASI-COMPACT QUASI-SEPARATED SCHEME COMMUTES WITH DIRECT SUMS OF BOUNDED BELOW COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction and the sheaf case are both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Cohomology of an object of the derived category at a scheme. Same chapter, the definition of H^p(X, K) for K in D(X_etale) and its computation from a resolution. Supplies the groups compared in the statement.
- Commutation of cohomology with direct sums for sheaves, and the bounded below hypothesis. Same chapter, the companion results stating that cohomology commutes with direct sums for sheaves and that the bounded below hypothesis controls the passage from sheaves to complexes. Supplies the two ingredients of the proof.

Source reference. Original lemma 07D9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7846–7846. Navigation pointer invoked at proof line 7846; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EZM: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7783–7783. Navigation pointer invoked at proof line 7849; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a quasi-compact and quasi-sepated scheme. Let $K_i \in D(X_{étale})$, $i \in I$ be a family of objects. Assume given $a \in \mathbf{Z}$ such that $H^n(K_i) = 0$ for $n < a$ and $i \in I$. Then $R\Gamma(X, \bigoplus_i K_i) = \bigoplus_i R\Gamma(X, K_i)$.*

Proof. We have to show that $H^p(X, \bigoplus_i K_i) = \bigoplus_i H^p(X, K_i)$ for all $p \in \mathbf{Z}$. Choose complexes $\mathcal{F}_i^\bullet$ representing K_i such that $\mathcal{F}_i^n = 0$ for $n < a$. The direct sum of the complexes $\mathcal{F}_i^\bullet$ represents the object $\bigoplus K_i$ by Injectives, Lemma 07D9. Since $\bigoplus \mathcal{F}^\bullet$ is the filtered colimit of the finite direct sums, the result follows from Lemma 0EZM. □